

Find the displacement over the given interval.

1. $v(t) = \cos\left(\frac{\pi}{2}t\right)$ on $[0, 3]$; $s(0) = 0$.

Determine the position function.

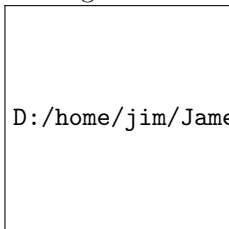
2. An object is moving at a velocity of $(t + 9)^{1/2}$ with an initial position of $s(0) = 20$ at time t .

Solve the problem.

3. A projectile is launched vertically from the ground at $t = 0$ where t is seconds, and its velocity in flight (in m/s) is given by $v(t) = 20 - 10t$.
 - a. Find the position at time t for $0 \leq t \leq 4$.
 - b. Find the total distance traveled from $t = 0$ to $t = 4$.

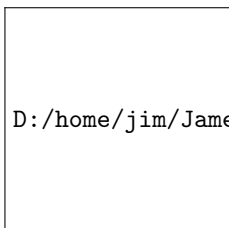
Use calculus to find the area of the region described.

4. The region bounded by $y = x$ and $y = x^2 - 4x$.



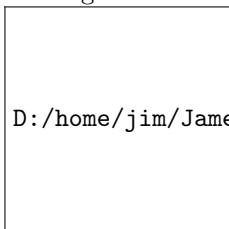
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5. The region bounded by $y = \frac{1}{2}x^2$ and $y = -x^2 + 6$.



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6. The region in the first quadrant bounded by $-x + 1$ and $x - 1$ from $0 \leq x \leq 2$.



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Use the general slicing method and calculus to find the volume of the solid.

7. The rectangular box with dimensions a height of 3, width of 4, and length of 10.

Let R be the region bounded by the following curves. Use the disk method to find the volume of the solid generated when R is revolved about the x -axis.

8. $y = \sqrt{x}$, $x = 0$, and $x = 9$.

9. $y = \sqrt{\sin 4x}$, $y = 0$, $0 \leq x \leq \frac{\pi}{4}$.

Find the length of the curve.

10. $y = 2x^{3/2}$ from $x = 0$ to $x = \frac{5}{4}$.

11. $y = \left(4 - x^{2/3}\right)^{3/2}$ from $x = 0$ to $x = 8$.

BONUS: Use calculus and the general slicing method to find a general formula for the volume of a pyramid with a square base.